

Dummy RAPM: Representing Low-Minute Players in Regularized Adjusted Plus-Minus

Kenny Watts, Jonathan Pipping-Gamón, and Abraham J. Wyner

The Wharton School, University of Pennsylvania

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Abstract

Regularized Adjusted Plus-Minus (RAPM) uses stint-level lineup indicators to estimate player contributions to scoring margin. When low-minute player columns are removed, their stints remain in the data, but the design matrix no longer represents the complete lineup. Dummy RAPM restores this information using five indicators for the number of excluded players on each lineup side. Across 17 NBA seasons, chronological validation selects a 10-minute-per-appearance threshold and a dummy-to-player penalty ratio of 2.2. On held-out March–April games, Dummy RAPM reduces mean season game-margin RMSE from 12.884 to 12.843 and achieves lower RMSE in 14 of 17 seasons. The average reduction is 0.041 points, or 0.30%. Although the improvement in game-level predictive accuracy is small, it is consistent: RAPM performs better when it records how many excluded players are on each side.

1 Introduction

Basic plus-minus measures a team’s score differential while a player is on the court but does not adjust for teammates or opponents. Adjusted Plus-Minus (APM), associated with the WINVAL work of Jeff Sagarin and Wayne Winston and described publicly by [Rosenbaum \(2004\)](#), addresses this limitation by regressing stint-level score differentials on player indicators.

APM estimates are unstable when players repeatedly share the floor with the same teammates and opponents or appear in few stints. Ridge regression stabilizes correlated designs by shrinking coefficients ([Hoerl and Kennard, 1970](#)). [Sill \(2010\)](#) applied this approach to APM and emphasized evaluation on future games, yielding Regularized Adjusted Plus-Minus (RAPM). Box Plus/Minus represents a separate, complementary line of work based on box-score and role information ([Myers, 2020](#)). Bayesian partial-pooling approaches provide a related response to weak identification. [Fearnhead and Taylor \(2011\)](#) estimated offensive and defensive abilities while borrowing information across seasons, and [Deshpande and Jensen \(2016\)](#) modeled changes in win probability using Bayesian linear regression.

Some RAPM implementations remove low-minute players before fitting. Removing these columns leaves their stints in the data but makes the lineup representation incomplete. The

model may then attribute omitted contributions to retained players whose indicators are correlated with the missing lineup information.

Dummy RAPM addresses this omission without estimating a separate coefficient for every low-minute player. It records how many excluded players appear on the home and away sides, then pools those count coefficients with ridge regularization. The question is simple: does restoring this small piece of lineup information improve prediction?

2 Model

2.1 Stint response and player design

A stint is an interval over which neither lineup changes. Let Δ_i be the change in home-minus-away score margin during stint i , and define exposure

$$q_i = \frac{P_{H,i} + P_{A,i}}{2},$$

where $P_{H,i}$ and $P_{A,i}$ count home and away possession-ending events. The response is

$$y_i = 100 \frac{\Delta_i}{q_i},$$

the home-team margin per 100 team possessions. Observation i receives weight q_i , so a longer stint contributes more to estimation. Scaling before estimation makes coefficients directly interpretable in points per 100 team possessions.

For retained player j , $x_{ij} = 1$ when the player is in the home lineup, $x_{ij} = -1$ when in the away lineup, and $x_{ij} = 0$ otherwise. Filtered RAPM estimates

$$(\hat{\alpha}, \hat{\beta}) = \arg \min_{\alpha, \beta} \left\{ \sum_{i=1}^n q_i (y_i - \alpha - \mathbf{x}_i^\top \beta)^2 + \lambda_P \|\beta\|_2^2 \right\}.$$

The intercept is not penalized, and predictors are not standardized. Ridge solutions are computed over a decreasing sequence of λ_P values using cyclic coordinate descent ([Friedman et al., 2010](#)).

2.2 Lineup-side count indicators

For $k = 1, \dots, 5$, define

$$d_{H,i}^{(k)} = \mathbb{1}\{\text{the home lineup in stint } i \text{ contains exactly } k \text{ excluded players}\},$$

$$d_{A,i}^{(k)} = \mathbb{1}\{\text{the away lineup in stint } i \text{ contains exactly } k \text{ excluded players}\}.$$

Lineups with no excluded players form the reference category. The augmented model therefore contains ten dummy regressors, five for each lineup side. Separate home and away coefficient

blocks allow the association for a given count to differ according to which side contains the excluded players.

Let γ_H and γ_A denote the two five-element dummy blocks. Dummy RAPM estimates

$$\arg \min_{\alpha, \beta, \gamma_H, \gamma_A} \sum_{i=1}^n q_i (y_i - \alpha - \mathbf{x}_i^\top \beta - \mathbf{d}_{H,i}^\top \gamma_H - \mathbf{d}_{A,i}^\top \gamma_A)^2 + \lambda_P \|\beta\|_2^2 + \lambda_D (\|\gamma_H\|_2^2 + \|\gamma_A\|_2^2).$$

The relative penalty $\rho = \lambda_D/\lambda_P$ is tuned from the data. A common penalty for the dummy coefficients allows them to be pooled more or less strongly than the player coefficients without requiring ten separate hyperparameters.

3 Data Construction

We use publicly accessible ESPN play-by-play and player box-score records distributed through the sportsdataverse project (Gilani, 2026). Starter flags identify each opening lineup, and recorded substitutions divide the game into stints with unchanged lineups. We retain a game only when both teams have five unique recorded starters and every modeled stint contains five unique home players and five unique away players. This removes 34 games, including 19 in 2019.

Stint scoring is reconstructed from changes in the reported home-minus-away scoreboard margin. This captures technical and other free throws even when the corresponding event is not marked as ending a possession. A scoring segment with no possession exposure is merged with the next positive-exposure stint, or with the preceding stint at the end of a game. We verify that the resulting stint margins sum to the official final margin. Within each game, we retain the contiguous sequence of periods beginning with period 1, preserving valid multi-overtime games while discarding disconnected records. Garbage-time and overtime possessions remain in the sample so that stint predictions aggregate to the full-game margin.

Player playing time is summarized by mean minutes per appearance within the relevant training window. This reflects a player’s role when active rather than the number of games he happened to play. As in Rosenbaum (2004), low-minute players are pooled rather than estimated individually, but we select the cutoff by chronological validation instead of fixing it at 250 total minutes across two seasons.

Table 1 reports the season-level sample. The analysis contains 20578 games and 519,745 positive-exposure stints. The “Out” column counts regular-season games in May–September, outside the analysis windows; these games may also appear among the starter exclusions.

4 Selection and Evaluation

The sample contains 17 seasons: 2007–2019 and 2022–2025. Seasons are labeled by the calendar year in which they end. The series begins in 2007 because starter metadata are

Table 1: Season-level analysis sample and chronological windows

Season	All	Oct–Dec	Jan–Feb	Oct–Feb	Mar–Apr	Out	Starter excl.	Stints	Players
2007	1214	443	404	847	362	0	5	28,257	454
2008	1217	448	411	859	358	0	0	27,243	451
2009	1228	471	409	880	348	0	0	26,348	445
2010	1225	470	409	879	346	0	0	26,835	441
2011	1226	479	408	887	339	0	0	27,583	452
2012	990	56	466	522	467	0	1	23,170	478
2013	1226	453	402	855	366	0	5	23,607	469
2014	1225	464	403	867	357	0	1	23,700	482
2015	1226	477	393	870	356	0	0	25,134	492
2016	1227	489	399	888	339	0	0	25,155	477
2017	1230	506	387	893	336	0	1	25,032	484
2018	1226	540	375	915	309	0	2	34,393	539
2019	1230	532	376	908	303	0	19	34,502	537
2022	1230	527	394	921	309	0	0	42,133	605
2023	1230	546	385	931	299	0	0	42,531	539
2024	1231	481	405	886	345	0	0	41,552	572
2025	1231	485	403	888	343	0	0	42,570	569

incomplete in earlier seasons. The disrupted 2019–20 season and the COVID-shortened 2020–21 season are excluded because neither supports the same October–April chronological split.

Within each season, October–December games form the inner training period, January–February games form the inner validation period, and March–April games form the outer test period. Outer training is exactly the union of the October–December and January–February windows. May–September games are in none of the four analysis sets. The windows are disjoint and strictly chronological.

The initial inner search considers mean-minutes-per-appearance thresholds

$$\{0, 5, 10, 15, 20, 25, 30\}$$

and dummy-to-player penalty ratios

$$\{0, 0.1, 0.25, 0.5, 1, 2\}.$$

Threshold zero retains every training-observed player and serves as the unfiltered RAPM benchmark. The cutoff determines which player columns are pooled and is chosen solely by predictive performance.

In the first stage, we evaluate every threshold-ratio pair in the initial grid. For each pair, we average the inner-validation RMSEs of Dummy RAPM and filtered RAPM, then average that score across seasons. The minimizing pair selects a threshold of 10 minutes per appearance. Holding that threshold fixed for both models, the second stage evaluates the dummy-to-player penalty ratio over

$$\rho \in \{1.0, 1.1, \dots, 2.5\}.$$

If the minimum occurs at a grid boundary, we extend the grid in increments of 0.1 until the selected value is interior. The checked grid runs from 1 to 2.5 and selects $\rho = 2.2$. Both stages minimize this model-averaged, season-averaged inner-validation RMSE. Exact ties prefer the lower minute threshold and then the larger dummy penalty. March–April results are not used for selection.

After selection, both models are refit using all games through February and evaluated once on March–April games. Player eligibility is recomputed using only the corresponding training period. For each season and training window, we construct one deterministic five-fold partition at the game level and use it for every threshold, model, and penalty ratio. Within each fit, λ_P minimizes cross-validated game-margin RMSE.

For game g , stint predictions are converted back to points and summed:

$$\hat{\Delta}_g = \sum_{i \in g} \hat{y}_i q_i / 100.$$

For season s , game-margin RMSE is

$$\text{RMSE}_s = \sqrt{\frac{1}{|\mathcal{T}_s|} \sum_{g \in \mathcal{T}_s} (\Delta_g - \hat{\Delta}_g)^2}.$$

Held-out predictive fit is summarized by

$$R_{\text{oos},s}^2 = 1 - \frac{\sum_{g \in \mathcal{T}_s} (\Delta_g - \hat{\Delta}_g)^2}{\sum_{g \in \mathcal{T}_s} (\Delta_g - \bar{\Delta}_{\mathcal{T}_s})^2},$$

where $\mathcal{T}_s$ is the March–April test set for season s and $\bar{\Delta}_{\mathcal{T}_s}$ is its mean observed game margin. The denominator is the test-set total sum of squares. Mean season RMSE and mean season R_{oos}^2 are arithmetic averages of these season-specific quantities, giving each season equal weight rather than pooling all test games.

The paired t -test on season-level RMSE differences is the primary inferential analysis. The Wilcoxon signed-rank and sign tests are rank-based and direction-only robustness checks; the unfiltered-benchmark and 2019-exclusion comparisons are secondary sensitivity analyses. These nominal tests treat seasons as independent observational units. Because adjacent seasons share players, coaches, and league conditions, their p -values are best interpreted as approximate across-season summaries.

Because predictions condition on the realized lineups in the held-out games, the target is lineup-conditioned final margin rather than a pregame forecast.

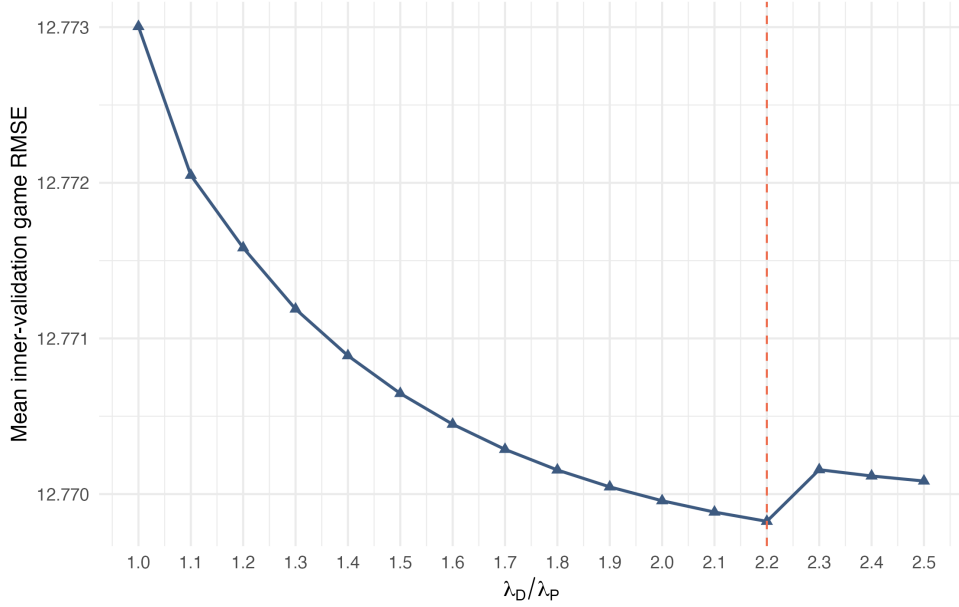


Figure 1: Mean inner-validation game-margin RMSE over the dummy-to-player penalty grid. The dashed line marks the selected ratio, 2.2.

5 Results

5.1 Held-out prediction

Table 2: Outer-test predictive performance across 17 seasons

Model	Mean season RMSE	Mean season R^2_{oos}	Lower-RMSE seasons
Dummy RAPM	12.843	0.174	14
Filtered RAPM	12.884	0.169	3

Dummy RAPM reduces mean season game-margin RMSE from 12.884 to 12.843. This reduction of 0.041 points, equivalent to 0.30% of the filtered RAPM RMSE, occurs in 14 of 17 seasons. Mean season R^2_{oos} rises from 0.169 to 0.174. Thus, the count indicators produce a small but consistent improvement in predictive accuracy across seasons.

The across-season 95% t -based confidence interval for the reduction in RMSE is 0.015 to 0.068 points. The paired t -test gives $t(16) = 3.310$ and $p = 0.0044$. As robustness checks, the Wilcoxon signed-rank test gives $W = 136.0$ and $p = 0.0052$. The sign test records lower RMSE in 14 of 17 seasons ($p = 0.0127$).

As a secondary sensitivity analysis, Dummy RAPM has lower RMSE in 10 of 13 seasons through 2019 and in 4 of 4 seasons from 2022 onward. Excluding 2019, the season with the most starter-data exclusions, leaves 16 seasons with an average RMSE reduction of 0.035 points ($p = 0.0084$).

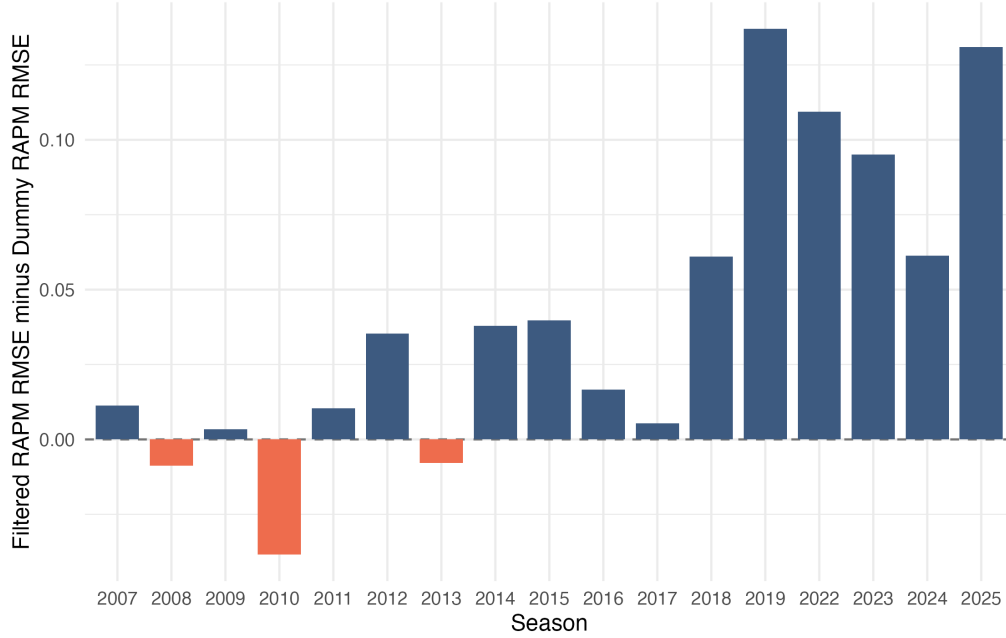


Figure 2: Outer-test game-margin RMSE difference by season. Positive values indicate lower RMSE for Dummy RAPM.

5.2 Filtering and representation

The unfiltered benchmark provides a second sensitivity analysis. Filtering low-minute players without adding count indicators does not improve RAPM. Relative to the unfiltered benchmark, the selected filtered model is worse by 0.014 RMSE on average ($p = 0.1125$). Dummy RAPM is better than that same benchmark by 0.027 RMSE ($p = 0.0445$). The improvement therefore appears to come from restoring information about lineup composition rather than from excluding low-minute player coefficients.

5.3 Dummy coefficients

Table 3: Selected outer-training dummy-category exposure across seasons

Side	Count	Stints	Exposure	Seasons	Seasons present
Home	0	353,707	1,303,375.0	17	2007–2019, 2022–2025
Home	1	20,810	63,360.5	17	2007–2019, 2022–2025
Home	2	2,533	8,203.5	17	2007–2019, 2022–2025
Home	3	691	2,857.5	13	2007, 2012–2019, 2022–2025
Home	4	259	1,101.0	7	2012, 2018, 2019, 2022–2025
Home	5	45	202.0	4	2022–2025
Away	0	351,609	1,297,976.0	17	2007–2019, 2022–2025
Away	1	22,418	67,852.0	17	2007–2019, 2022–2025
Away	2	2,935	9,255.5	17	2007–2019, 2022–2025
Away	3	769	2,738.5	14	2007, 2010, 2012–2019, 2022–2025
Away	4	269	1,096.5	8	2012, 2016, 2018, 2019, 2022–2025
Away	5	45	181.0	4	2022–2025

Table 3 reports how often each home and away count occurs in the selected outer-training designs. Exposure is the sum of $q_i = (P_{H,i} + P_{A,i})/2$ over contributing stints. The season lists distinguish structural zeros caused by an absent category from coefficients merely shrunk close to zero.

Table 4: Selected dummy coefficients across seasons

Lineup side	Low-minute players	Mean coefficient	Directional seasons
Home	1	-1.51	17 negative
Home	2	-0.61	17 negative
Home	3	-0.36	13 negative, 4 zero
Home	4	-0.07	5 negative, 2 positive, 10 zero
Home	5	-0.04	4 negative, 13 zero
Away	1	0.78	3 negative, 14 positive
Away	2	0.24	7 negative, 10 positive
Away	3	-0.05	12 negative, 2 positive, 3 zero
Away	4	-0.07	5 negative, 3 positive, 9 zero
Away	5	0.02	1 negative, 3 positive, 13 zero

The dominant coefficient pattern is straightforward. Conditional on the retained-player indicators, a lineup with one excluded home player is associated with -1.51 home-margin points per 100 possessions. One excluded away player is associated with 0.78 home-margin points. Both signs are consistent with the same basketball interpretation: the side using a

low-minute player tends to be weaker. The home coefficient is negative in 17 seasons, and the away coefficient is positive in 14.

These are lineup associations, not ratings of individual excluded players. Score state, injuries, matchups, and schedule conditions all affect when low-minute lineups appear. Counts of four and five occur in only a few seasons; when a category is absent, its coefficient is structurally zero. The clearest recurring pattern is concentrated in the one- and two-player categories. The intervals in Figure 3 summarize variation across seasons.

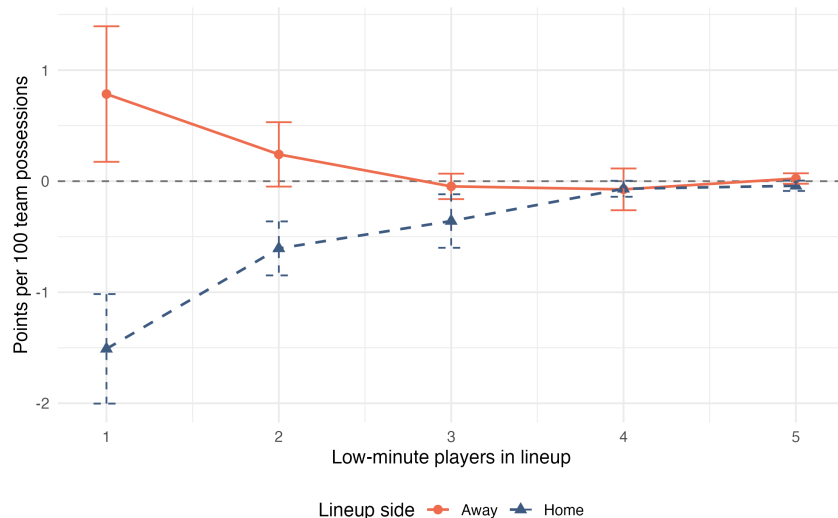


Figure 3: Mean selected home and away dummy coefficients with 95% cross-season t intervals. Away coefficients retain the home-margin orientation.

6 Discussion

Why does this modest modification improve prediction so consistently? Most of the predictive signal is already carried by the retained-player indicators. Dropping low-minute columns discards only a limited amount of lineup information, so a large reduction in RMSE should not be expected. Nevertheless, the omitted information is predictive. Ten count indicators recover enough of it to improve prediction in 14 of 17 seasons.

The comparison with the unfiltered benchmark sharpens the point. Filtering players while omitting their lineup counts makes the model slightly worse. Recording those counts makes it slightly better. RAPM does not need a separate, noisy coefficient for every low-minute player, but it benefits from knowing how many such players are on the floor.

The selected ratio, $\rho = 2.2$, places 2.2 times as much penalty weight on the squared count coefficients as on the squared player coefficients. Because predictors are not standardized and occur at different frequencies, ρ is not an effective shrinkage factor and should not be read as the literal relative contraction of the fitted coefficients. Nearby ratios perform nearly identically, so the precise value of 2.2 is less important than the broader result: the data favor a distinct, relatively strong penalty for the count coefficients rather than their exclusion.

The remaining weaknesses come mostly from the data. Lineups are reconstructed from play-by-play and starter records rather than an official shift chart. Starter-data exclusions are concentrated in 2019, although the result survives without that season. Categories with four or five low-minute players are rare, so they contribute little to the substantive pattern.

7 Conclusion

When low-minute players are filtered out, they disappear from the design matrix but not from the game. Five home and five away count indicators preserve that lineup information. Across 17 seasons, this representation improves game-margin RMSE in 14 seasons. RAPM can therefore pool low-minute players without discarding information about their presence on the floor.

Reproducibility

Code and instructions for reproducing the data construction, model fitting, tables, figures, and manuscript are available at <https://github.com/whartonsabi/dummy-rapm>.

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References

- Deshpande, S. K. and Jensen, S. T. (2016). Estimating an NBA player’s impact on his team’s chances of winning. *Journal of Quantitative Analysis in Sports*, 12:51–72.
- Fearnhead, P. and Taylor, B. M. (2011). On estimating the ability of NBA players. *Journal of Quantitative Analysis in Sports*, 7(3).
- Friedman, J., Hastie, T., and Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1):1–22.
- Gilani, S. (2026). *hoopR: Access Men’s Basketball Play-by-Play Data*. R package version 3.1.0.
- Hoerl, A. E. and Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67.
- Myers, D. (2020). Introducing Box Plus/Minus 2.0.
- Rosenbaum, D. T. (2004). Measuring how NBA players help their teams win.
- Sill, J. (2010). Improved NBA adjusted plus-minus using regularization and out-of-sample testing.